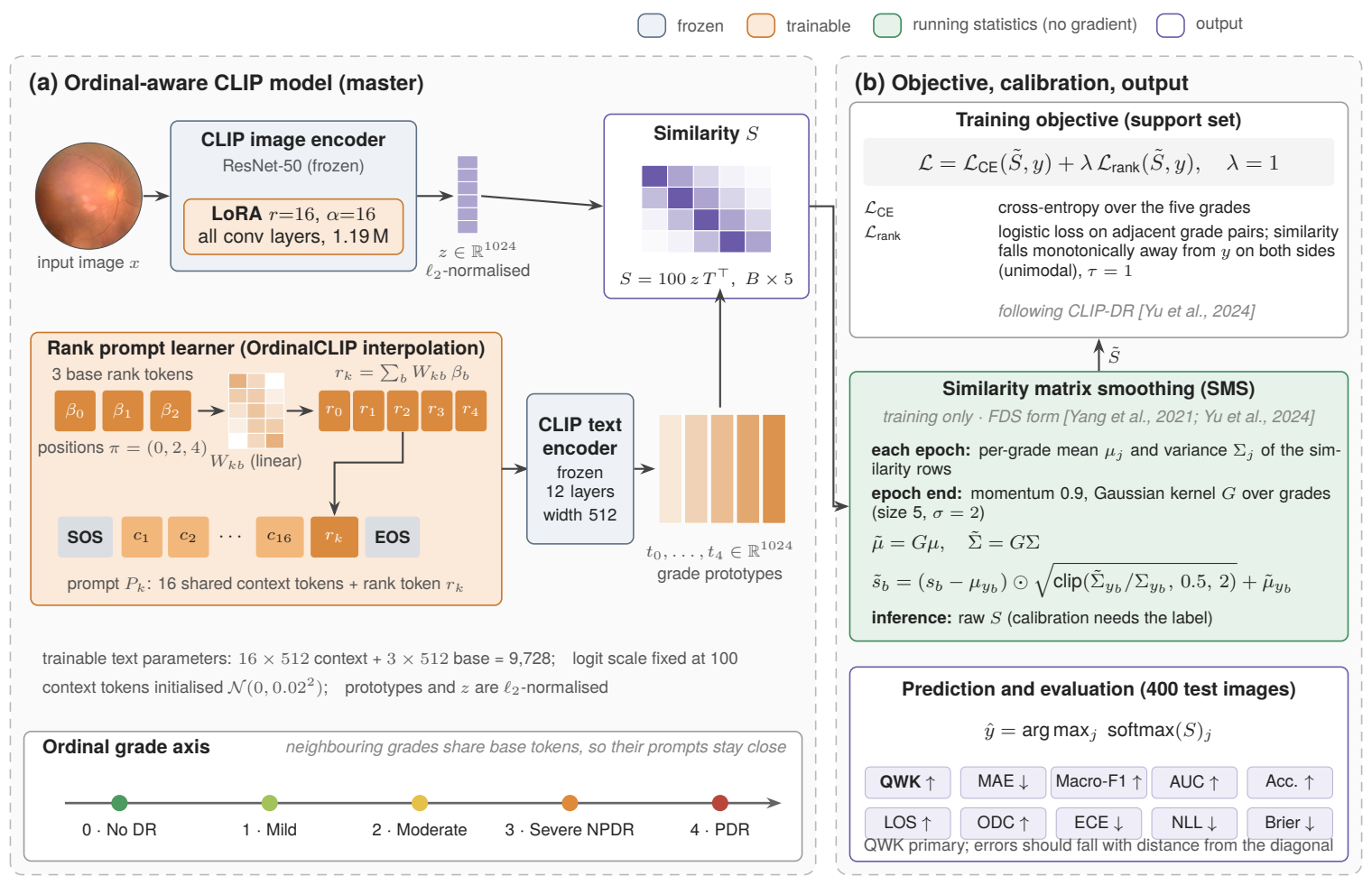

frozen
trainable
running statistics (no gradient)
output
(a) Ordinal-aware CLIP model (master)
input image x
CLIP image encoder
ResNet-50 (frozen)
LoRA r=16, α=16
all conv layers, 1.19 M
$z \in \mathbb{R}^{1024}$
ℓ_2-normalised
Similarity S
$S = 100\, z\, T^\top,\ B \times 5$
Rank prompt learner (OrdinalCLIP interpolation)
3 base rank tokens
β_0 β_1 β_2
positions $\pi = (0, 2, 4)$
W_{kb} (linear)
$r_k = \sum_b W_{kb}\, \beta_b$
r_0 r_1 r_2 r_3 r_4
SOS c_1 c_2 $\cdots$ c_{16} r_k EOS
prompt P_k: 16 shared context tokens + rank token r_k
CLIP text encoder
frozen
12 layers
width 512
$t_0, \ldots, t_4 \in \mathbb{R}^{1024}$
grade prototypes
trainable text parameters: 16×512 context + 3×512 base = 9,728; logit scale fixed at 100
context tokens initialised $\mathcal{N}(0, 0.02^2)$; prototypes and z are ℓ_2-normalised
Ordinal grade axis
neighbouring grades share base tokens, so their prompts stay close
0 · No DR
1 · Mild
2 · Moderate
3 · Severe NPDR
4 · PDR
(b) Objective, calibration, output
Training objective (support set)
$\mathcal{L} = \mathcal{L}_{\text{CE}}(\tilde{S}, y) + \lambda\, \mathcal{L}_{\text{rank}}(\tilde{S}, y), \quad \lambda = 1$
$\mathcal{L}_{\text{CE}}$ cross-entropy over the five grades
$\mathcal{L}_{\text{rank}}$ logistic loss on adjacent grade pairs; similarity falls monotonically away from y on both sides (unimodal), $\tau = 1$
following CLIP-DR [Yu et al., 2024]
$\tilde{S}$
Similarity matrix smoothing (SMS)
training only · FDS form [Yang et al., 2021; Yu et al., 2024]
each epoch: per-grade mean μ_j and variance Σ_j of the similarity rows
epoch end: momentum 0.9, Gaussian kernel G over grades (size 5, $\sigma = 2$)
$\tilde{\mu} = G\mu, \quad \tilde{\Sigma} = G\Sigma$
$\tilde{s}_b = (s_b - \mu_{y_b}) \odot \sqrt{\text{clip}(\tilde{\Sigma}_{y_b}/\Sigma_{y_b},\, 0.5,\, 2)} + \tilde{\mu}_{y_b}$
inference: raw S (calibration needs the label)
Prediction and evaluation (400 test images)
$\hat{y} = \arg\max_j\ \text{softmax}(S)_j$
QWK ↑
MAE ↓
Macro-F1 ↑
AUC ↑
Acc. ↑
LOS ↑
ODC ↑
ECE ↓
NLL ↓
Brier ↓
QWK primary; errors should fall with distance from the diagonal